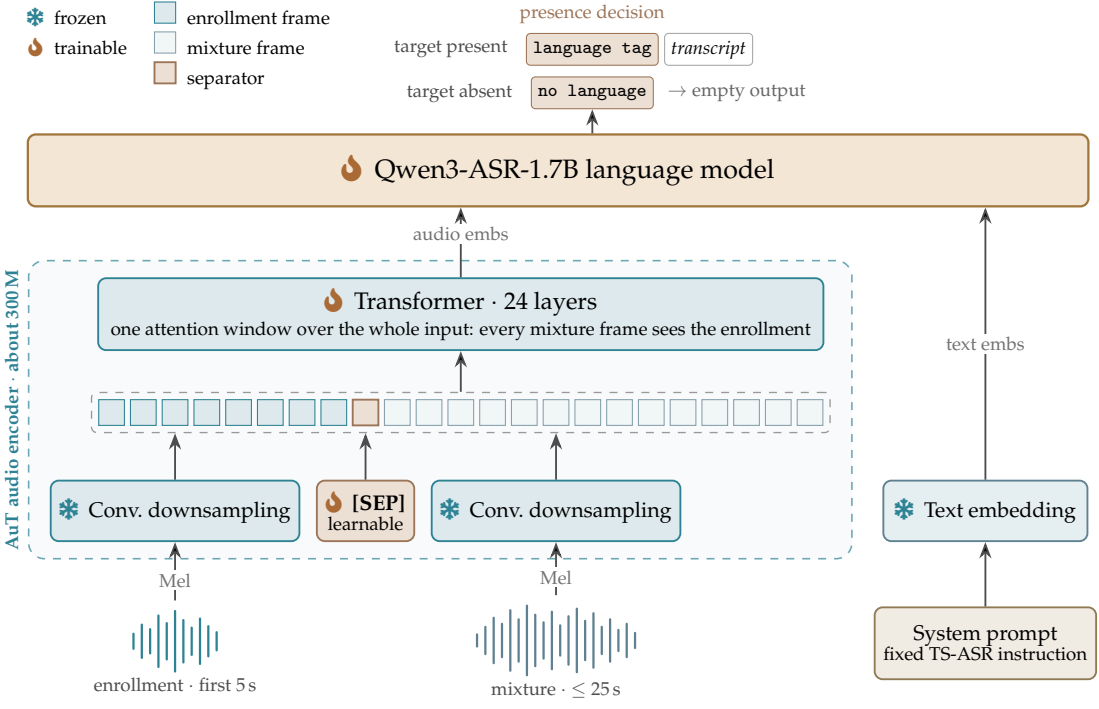

frozen
trainable
enrollment frame
mixture frame
separator
presence decision
target present
language tag
transcript
target absent
no language
→ empty output
Qwen3-ASR-1.7B language model
audio embs
text embs
AuT audio encoder · about 300 M
Transformer · 24 layers
one attention window over the whole input: every mixture frame sees the enrollment
Conv. downsampling
[SEP]
learnable
Conv. downsampling
Text embedding
Mel
Mel
enrollment · first 5 s
mixture · ≤ 25 s
System prompt
fixed TS-ASR instruction